

Assignment 4

Connecting and Testing Peripherals Practical

Task (1):

Connect a USB webcam to your computer, install any required drivers, and verify that the webcam works by opening the default camera app and capturing a photo.

Ans.

❖ USB Webcam Installation and Verification:

Step 1: Connect the Webcam

- Plug the USB webcam into an available USB port on your computer.
- Wait for Windows to detect the new device.

Step 2: Install Drivers

- Windows 10/11 will usually install the required drivers automatically.
- If prompted, allow the installation to complete.

Step 3: Verify the Webcam in Device Manager

- Press Windows + X and select Device Manager.
- Expand Cameras or Imaging Devices.

Step 4: Open the Camera App

- Press the Windows key and search for Camera.
- Open the Camera app.
- If asked, click Allow to give the app permission to access the camera.

Step 5: Test the Webcam

- Check that the live video from the webcam is displayed.
- Click the Camera button to capture a photo.

Step 6: Verify the Photo

- Open the captured image from the Camera app's gallery or the Pictures → Camera Roll folder.
- Confirm that the photo is clear and saved successfully.

Task (2):

Set up a local printer (USB or network), install the correct driver, and print a test page to confirm successful installation.

Ans.

Local Printer Setup and Test Page:

Step 1: Connect the Printer

- **USB Printer:** Connect the printer to your computer using a USB cable and turn it on.
- **Network Printer:** Connect the printer to the same Wi-Fi or Ethernet network as your computer and power it on.

Step 2: Install the Printer Driver

- Windows will usually detect the printer and install the required driver automatically.
- If it does not, install the correct driver from the printer manufacturer's installation disc or website.

Step 3: Add the Printer

- Open Settings → Devices → Printers & scanners.
- Click Add a printer or scanner.
- Select your printer from the list.
- Click Add device and wait for the installation to finish.

Step 4: Verify the Installation

- Open Control Panel → Devices and Printers.
- Confirm that the printer appears in the list and its status is Ready.

Step 5: Print a Test Page

- Right-click the installed printer.
- Select Printer properties.
- Click Print Test Page.

Step 6: Confirm Successful Installation

- Check that the test page prints correctly with text, graphics, and printer information.

Task (3):

Simulate a connectivity issue by unplugging the printer's USB cable, then use Windows Device Manager to identify and resolve the 'printer not detected' problem. Hint: Check for error icons and use the 'Scan for hardware changes' option after reconnecting.

Ans.

Step 1: Disconnect the Printer

- Turn on the computer and printer.
- Unplug the printer's USB cable from the computer.
- Wait a few seconds.

Step 2: Open Device Manager

- Press Windows + X and select Device Manager.
- Expand Printers, Print queues, or Universal Serial Bus controllers.
- Observe whether the printer disappears or shows a yellow warning icon.

Step 3: Reconnect the Printer

- Plug the USB cable back into the same or a different USB port.
- Wait for Windows to detect the printer.

Step 4: Scan for Hardware Changes

- In Device Manager, click the Action menu.
- Select Scan for hardware changes.

Step 5: Check Device Status

- Verify that the printer reappears under Printers or Print queues.
- Ensure there are no yellow warning icons.
- If a warning icon is present, right-click the printer and select Update driver or Uninstall device, then scan for hardware changes again.

Step 6: Test the Printer

- Open Settings → Devices → Printers & scanners.
- Confirm the printer status is Ready.
- Print a Test Page from Printer properties to verify that the printer is working correctly.

Task (4):

A user reports that their webcam video is not displaying in Zoom, but it works in the default camera app. Troubleshoot and resolve the issue so the webcam works in both applications. Hint: Check if another app is using the webcam or if Zoom permissions are enabled.

Ans.

Step 1: Close Other Applications:

- Close the Camera app and any other applications that may be using the webcam (Microsoft Teams, Skype, Google Meet, OBS, etc.).
- Reopen Zoom and test the camera.

Step 2: Check Zoom Camera Settings

- Open Zoom.
- Click Settings → Video.
- Under Camera, select the correct webcam from the drop-down list.

Step 3: Check Windows Camera Permissions

- Open Settings → Privacy → Camera (Windows 10) or Settings → Privacy & Security → Camera (Windows 11).

Step 4: Restart Zoom

- Completely exit Zoom (including from the system tray).
- Reopen Zoom and join or start a meeting to test the webcam.

Step 5: Verify Device Manager

- Press Windows + X → Device Manager.
- Expand Cameras.
- Confirm the webcam is listed without a yellow warning icon.
- If necessary, update or reinstall the webcam driver.

Step 6: Restart the Computer

- Restart the PC to release any application that may still be using the webcam.
- Open Zoom again and test the camera.

Step 7: Update Zoom

- Ensure Zoom is updated to the latest version.
- Restart Zoom after updating.

Resolution

- After closing other applications, enabling camera permissions for desktop apps, selecting the correct webcam in Zoom, and restarting Zoom, the webcam should display correctly in both the Windows Camera app and Zoom.
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